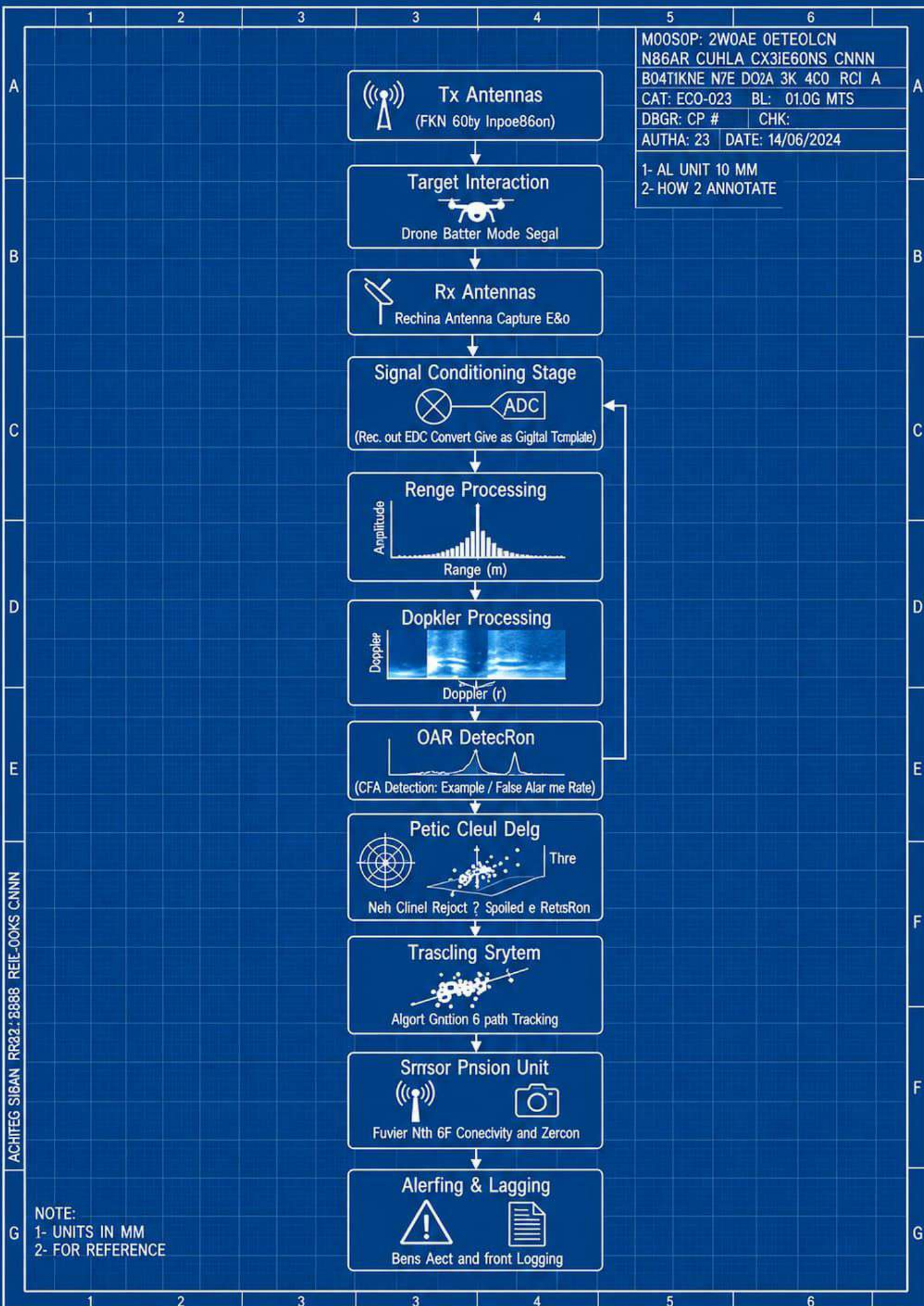
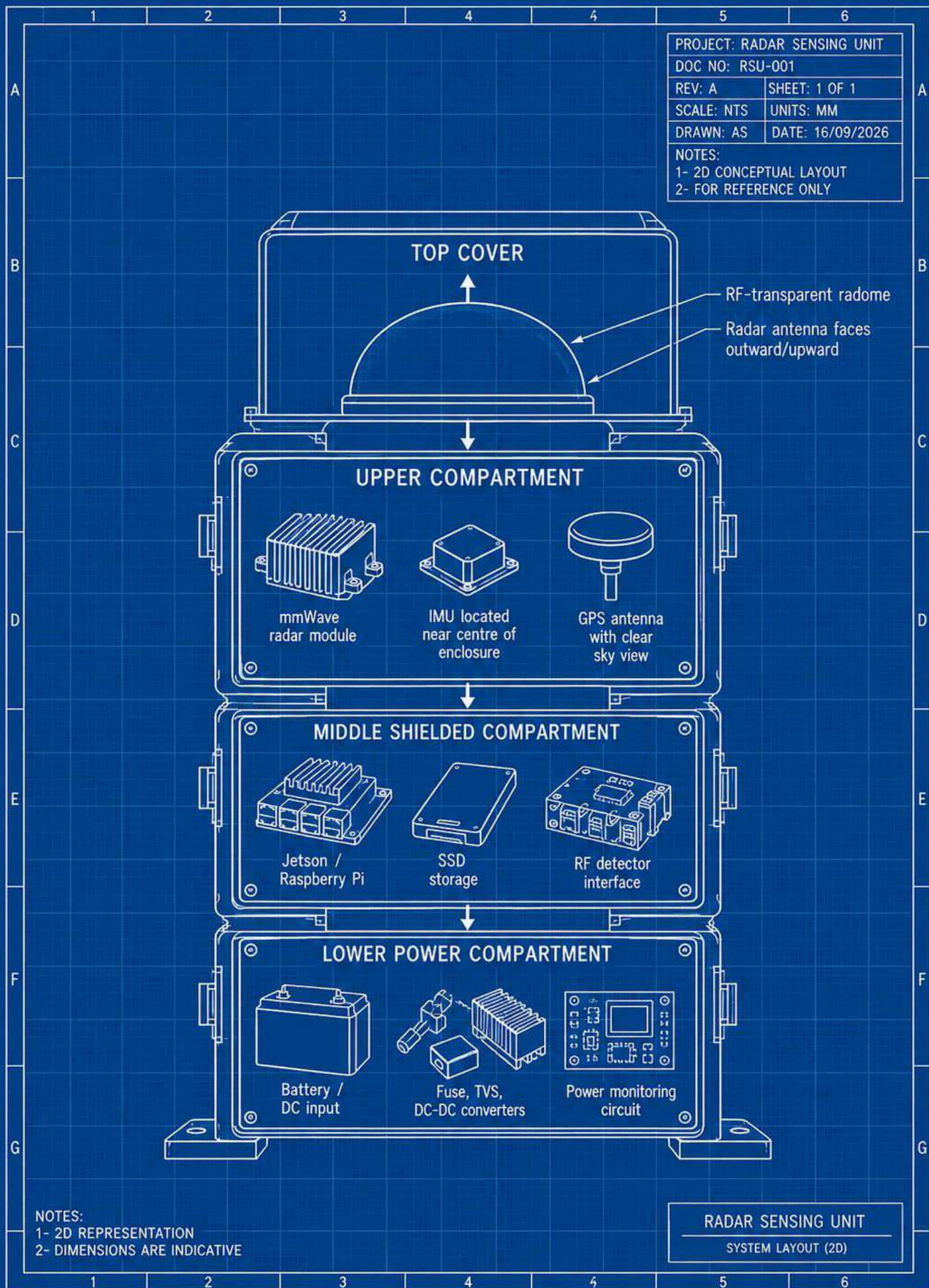
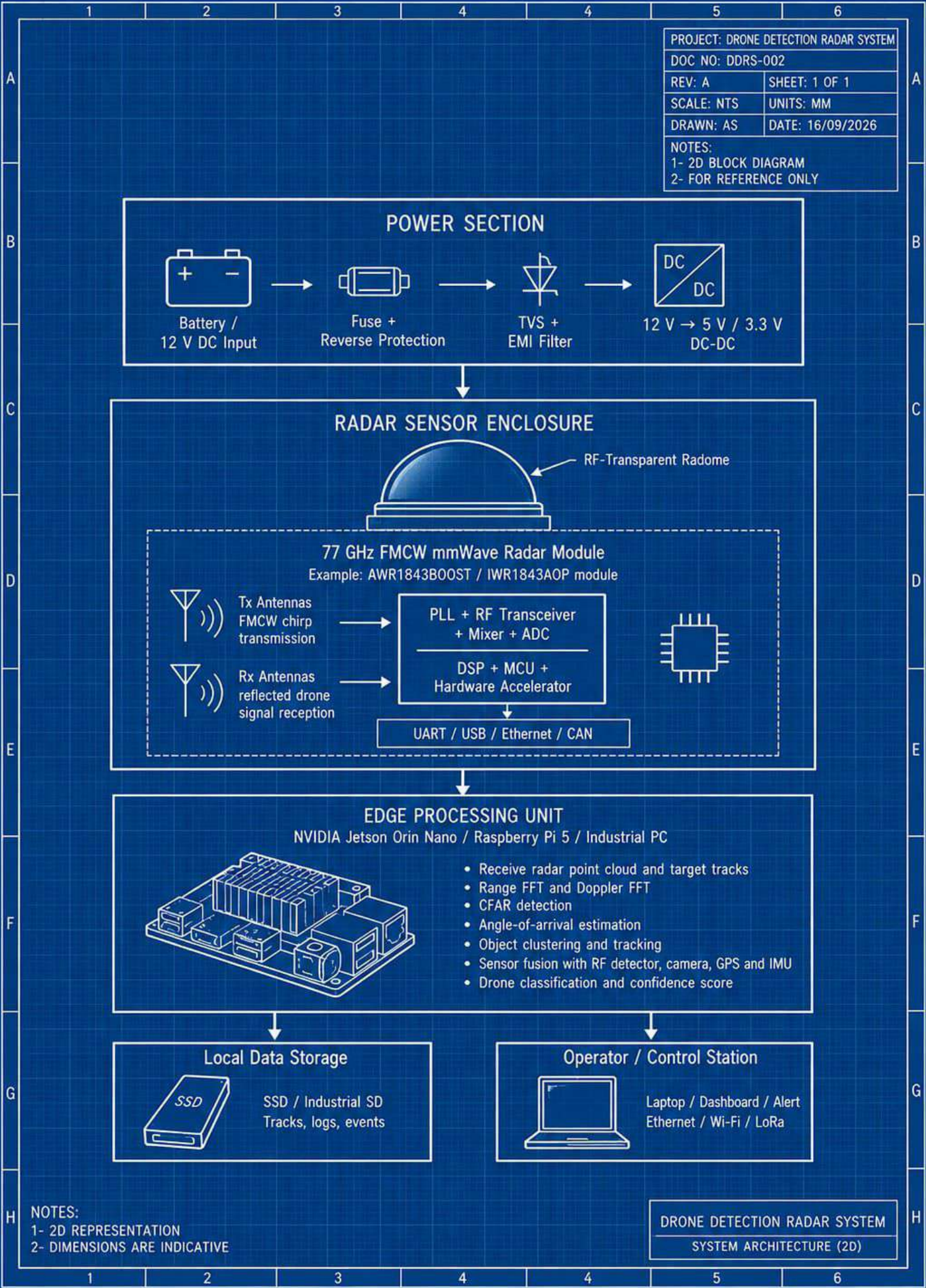
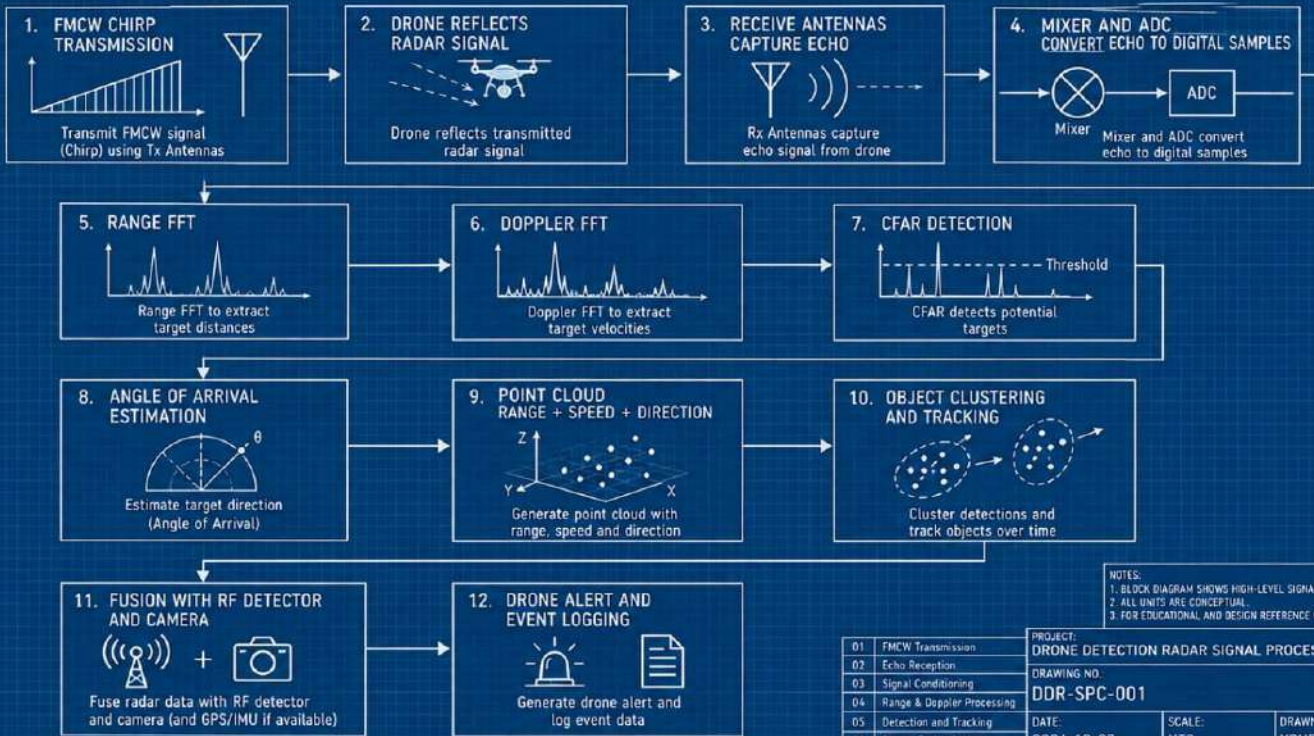


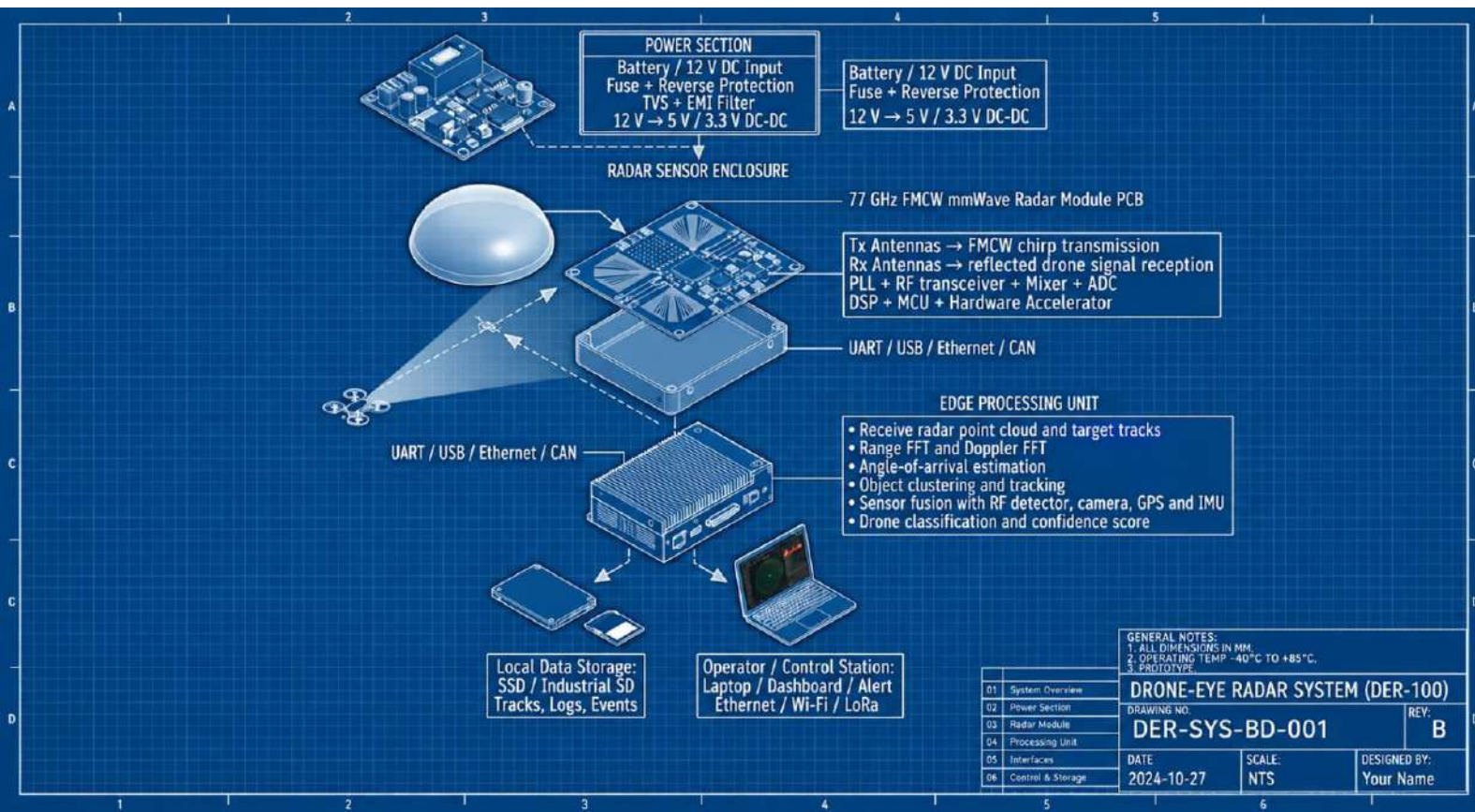






DRONE DETECTION RADAR SIGNAL PROCESSING CHAIN





Hardware and Sensors Research sheet by Aman Singh

Hardware and sensor information for the counter-UAS SIH prototype, structured for direct use in the report or presentation.

1. Proposed Sensor Suite

Sensor	Purpose	Keep or not
RF detection module	Detect drone remote-control and video-transmission signals; extract frequency, signal strength and direction	Must keep
Radar module	Measure drone range, speed, altitude and position	Keep as primary sensor
EO camera	Visual identification and confirmation in daylight	Must keep
IR/thermal camera	Detection at night, in low light and via thermal signature	Useful if budget allows
Acoustic microphone array	Detect drone motor/propeller sound	Optional; accuracy drops in high wind
GPS/GNSS module	System location and time synchronization	Useful for mobile/deployed systems
IMU: accelerometer + gyroscope	Measure tilt, vibration and orientation of antenna/camera platform	Must keep
Barometric pressure sensor	Estimate altitude and atmospheric pressure	Important for high-altitude project
Temperature and humidity sensor	Monitor electronics health and environmental conditions	Must keep
Current/voltage sensor	Monitor battery, power supply and abnormal current	Must keep

Counter-UAS systems typically prioritize RF, radar and visible-light sensors; fusing EO/IR with RF/radar data improves target confirmation and tracking.

Recommended minimum configuration

For the prototype, this configuration is sufficient:

1. RF detector
2. Short-range radar or radar simulator
3. Daylight camera
4. IMU
5. Temperature, humidity and pressure sensors
6. Voltage/current monitoring circuit
7. GPS module, if the system is mobile

Keep the acoustic sensor optional in the first prototype, because wind, vehicles and birds can cause false alarms. Add a thermal camera only if night operation is required and budget allows.

2. What Data Will Come From Each Sensor?

Each sensor will produce the following types of data:

Sensor	Data
RF detector	Frequency, bandwidth, RSSI/signal strength, modulation features, direction
Radar	Range, azimuth, elevation, radial velocity, target track
EO camera	Image frames, bounding box, object class, confidence score
IR camera	Thermal frames and heat signature
Acoustic array	Sound level, frequency spectrum, direction of arrival
GPS	Latitude, longitude, altitude and time
IMU	Acceleration, angular velocity and platform orientation
Pressure sensor	Atmospheric pressure and estimated altitude
Temperature/humidity	Internal temperature, external temperature and relative humidity
Power monitor	Voltage, current, power consumption and battery state

Importantly, instead of streaming full raw video continuously to the cloud, process the camera feed on an edge computer. Initial detection from the camera happens on the local processor, and only necessary metadata or alerts are forwarded.

3. Hardware Architecture and Data Flow

Sensor Interface Board → Edge Computer / AI Processor → Sensor fusion, drone classification, track generation, threat scoring → Local display / operator console, local

storage, secure communication link.

Where will data be collected?

Collect data at three levels:

8. Sensor level: Raw data from each sensor arrives at a microcontroller or interface board.
9. Edge level: A Raspberry Pi, NVIDIA Jetson or industrial embedded computer timestamps and combines sensor data.
10. Control-station level: Only alerts, target tracks and necessary image/operator data are sent to a command laptop or monitoring station.

Where will the data go?

- Real-time detection data: RAM/processing pipeline of the edge computer.
- Target tracks and alerts: local database or SD card/SSD.
- Selected images and event logs: encrypted local storage.
- Operator dashboard: via Ethernet, Wi-Fi, private 4G/5G or LoRa, depending on range and security needs.
- Internet/cloud: only if permitted; do not make cloud the default for sensitive operational data.

For each event, save this information:

- Timestamp
- Sensor ID
- Target ID
- Latitude, longitude, altitude
- Range and speed
- RF characteristics
- Camera image reference
- Classification result
- Confidence score
- System health
- Operator action

4. Electrical Components and Hardware Upgrades

Core electrical components

- Main processor: NVIDIA Jetson Orin Nano or similar edge-AI computer.
- Microcontroller: STM32, ESP32 or industrial MCU for sensor reading and health monitoring.
- Power source: Rechargeable battery pack or regulated DC supply.

- DC-DC converters: Separate regulated rails such as 12 V, 5 V and 3.3 V.
- Power Management Unit: fuse, reverse-polarity protection, over-voltage protection and under-voltage cutoff.
- Communication interfaces: USB 3.0, Ethernet, UART, SPI, I²C, CAN or RS-485.
- Storage: Industrial-grade microSD/SSD; SSD is better for continuous video.
- Real-time clock: For accurate event timestamps.
- Cooling system: Heat sink, heat spreader and low-power fan or conduction cooling.
- EMI filters: To reduce RF interference and power-line noise.
- Surge and ESD protection: TVS diodes and grounding for outdoor cables and connectors.

Recommended upgrades

1. Use locking industrial connectors instead of consumer-grade wires.
2. Add a separate fuse for each important power branch.
3. Measure both battery voltage and current.
4. Place the processor and RF section in separate compartments.
5. Design a safe degraded mode for sensor-failure conditions.
6. Add connector labels and test points on all boards.
7. Do not always store raw video; use event-triggered recording.
8. Maintain primary and backup communication channels.
9. Add a hardware watchdog so the software restarts automatically if it hangs.
10. Mount all PCBs on vibration-resistant standoffs.

5. High-Altitude Protection

At high altitude, lower pressure can reduce cooling efficiency and increase stress on electrical insulation. Therefore, high-altitude design must focus on component derating, adequate insulation spacing and conduction cooling.

Mechanical protection

- Sealed aluminium or reinforced polymer enclosure.
- Shock-absorbing rubber mounts.
- Internal PCB supports and cable strain relief.
- Locking connectors.
- Replaceable dust filter, if a fan is used.
- Drain/breather design to reduce condensation.

Environmental protection

- Target IP65 or IP67 for the outdoor enclosure.

- Apply conformal coating on PCBs.
- Use corrosion-resistant screws and brackets.
- Install cable glands at cable entries.
- Add sealing boots on connector joints.
- Perform high-altitude cold-start and low-pressure testing.

IP ratings describe protection against dust and water, while standards like MIL-STD-810 define environmental tests for temperature, altitude, vibration, shock, humidity and dust; they complement each other and are not substitutes.

Conformal coating

PCB coating options:

Coating	Advantages	Limitations
Acrylic	Cheap, easy to repair	Moderate chemical and moisture protection
Silicone	Flexible, good for vibration and temperature variation	Difficult to repair/debond
Polyurethane	Strong chemical and moisture resistance	Rework is difficult
Parylene	Excellent uniform protection	Expensive and requires specialised process

For the prototype, acrylic or silicone conformal coating is practical. Do not coat RF connectors, switches, exposed sensor openings, heat-generating parts or test points. Clean and fully dry the PCB before coating.

6. Camouflage and Low-Visibility Design

Camouflage is not just painting the system green. You should control visual, thermal and electromagnetic signatures.

Visual camouflage

- Use matte, non-reflective finishes.
- Choose olive, sand or grey colours depending on the environment.
- Cover glossy metal surfaces.
- Do not block camera lenses or radar apertures with paint.
- Make the enclosure outline irregular if possible.
- Cover cables and bright indicators.
- Keep only operator-facing status LEDs; make them dim or shielded.

Thermal signature reduction

- Do not let heat from the processor and power converters concentrate in a single hot spot.
- Use heat spreaders.
- Do not direct fan exhaust straight upwards.
- Use thermal insulation and passive heat sinking.
- Avoid unnecessary hot surfaces on the outside of the enclosure.

RF/EMI considerations

- Do not place metal camouflage nets in front of RF antennas.
- Use RF-transparent radomes for radar and RF antennas.
- Put internal processors and power electronics in shielded compartments.
- Design shielding with proper single-point grounding.
- Ensure camouflage materials do not obstruct sensor performance or radio communication.

Important: Present camouflage as a way to protect equipment from weather, glare and visual clutter, not as an offensive feature to avoid detection or identification. Your SIH prototype should be limited to defensive monitoring and authorized testing.

7. Software and Sensor Fusion

Implement this processing pipeline on the edge computer:

1. Sensor data acquisition.
2. Timestamp and synchronization.
3. Noise filtering.
4. Individual sensor detection.
5. Radar/RF/camera track association.
6. Target classification.
7. Confidence-score calculation.
8. Operator alert.
9. Event logging.

Example: Radar detects object + RF sensor detects drone-like signal + Camera confirms flying object = High-confidence drone alert.

If only camera detection is available without RF and radar confirmation, show the alert as “low confidence”. This reduces false alarms from birds, kites and other moving objects.

8. Sensors Not to Use in the First Prototype

In the first phase, keep the following sensors as optional or future upgrades:

- Expensive long-range 3D radar.
- Full thermal imaging system, if night operation is not required.
- Acoustic array, if the testing environment is noisy.
- Unnecessary chemical, gas or air-quality sensors.
- Multiple overlapping temperature sensors.
- Cloud-only data logging.
- Unverified military-grade modules with unclear interface and power requirements.

Build a minimum viable system first, then add sensors based on measured test results. This keeps cost, power consumption and integration complexity under control.

9. Suggested Hardware Update Text for Your Report

Hardware and Sensor Update:

The proposed system uses a layered sensing architecture consisting of RF detection, radar, EO camera, IMU, GPS, barometric pressure, temperature, humidity and power-monitoring sensors. RF and radar sensors provide early detection, while EO/IR sensors provide visual confirmation. Sensor data is timestamped and transferred to an edge-computing unit for filtering, sensor fusion, classification and track generation. Only essential alerts, target tracks and selected images are transmitted to the operator console, while event logs are stored locally. The electronics are protected using a sealed IP65/IP67 enclosure, conformal-coated PCBs, EMI shielding, surge protection, vibration mounts and regulated power supplies. A matte, non-reflective and RF-transparent external finish is proposed to reduce glare without obstructing antennas or cameras.

A multi-sensor fusion approach based on RF, radar and EO/IR is the most defensible technical choice for your project; keeping acoustic sensing optional and prioritizing edge processing will make the prototype more practical and reliable.